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Beyond Proximity: How Much of a 3D Scene Should a Language Model Read? A Controlled Study of Hierarchical Context Reduction

ANDREY KUZNETSOV[1], MAHMOUD MOUSATAT[1], AHMAD TAHA[1], ALEKSANDR MEDVEDEV[1], LEO VESIN[1], TELMAN NURUZOV[1], AND NIKITA SHANKIN[1]

[1]Independent researchers, Beyond Proximity project (e-mail: medvedguk@gmail.com)

Corresponding author: Aleksandr Medvedev (e-mail: medvedguk@gmail.com).

ABSTRACT Language interfaces to digital twins and robot maps increasingly hand the final decision to a large language model (LLM) that reads serialized scene records. We ask how much of a 3D scene such a model must read, and whether a spatial hierarchy selects that evidence better than a flat cheap-first cascade. We formalize a recall-controlled traversal, in which a cached proxy ranks branches, an adaptive probability-mass gate keeps them, and a global safety shortlist protects the proxy's best records, and evaluate it under same-scorer, matched-budget controls on 125 labeled queries over five rendered scenes, on 155,535 ReferIt3D expressions from 1,436 ScanNet scenes, and on ScanNet and Replica object maps. Four findings emerge. First, a cheap BGE proxy in front of Qwen2.5-7B halves the model's input (2,236 to 1,090 tokens) and latency (5.7 to 2.9 s) at a hit@3 cost of 0.027 (95% CI $[-0.067, 0]$), and giving the model three retained records instead of all of them raises its answer accuracy from 0.613 to 0.733 (+0.120, 95% CI [0.013, 0.227]). Second, at a matched candidate budget, hierarchical selection is not better than flat top-N selection, and when proxy and scorer coincide it places the target at the same rank as flat top-k on every held-out query. Third, the safety shortlist makes retrieval insensitive to the hierarchy: manual zones and pose-based and raw RGB-D hierarchies are statistically indistinguishable, whereas a fixed two-branch gate loses up to 0.096 hit@3. Fourth, at 155,535 records a scene hierarchy without localization at best ties flat search, although 96.9% of flat misses there come from other scenes. We release all traces, and we document and correct an orientation error in a Replica ground-truth box import that had understated predicted-map accuracy.

INDEX TERMS 3D scene graphs, hierarchical retrieval, context reduction, retrieval cascades, large language models, open-vocabulary grounding, RGB-D scene understanding, approximate nearest-neighbor search.

I. INTRODUCTION

ASKING a building, a museum, or a robot map a question in natural language—"where can I sit near reception?", "which exit is closest to the stage?"—is becoming a routine interface to digital twins and mobile robots. Current systems answer such queries in two stages. A perception stage turns posed RGB-D captures into records: object instances with boxes and captions [1], [2], hierarchical floor–room–object graphs [3], [4], or language-embedded fields [5], [6]. A reasoning stage then serializes these records into the prompt of an LLM or a vision–language model, which selects the answer [7], [8].

The reasoning stage is where cost and failure concentrate. Its price, latency, and memory grow with the number of serialized records, and its accuracy can drop when relevant evidence is buried in a long context [9]. A single room already yields tens of records; a building yields thousands. It is therefore natural to prune the scene before the model reads it, and spatial structure is an intuitive way to do so: "near reception" already rules out most of a building. Scene-graph systems follow this intuition. SayPlan lets an LLM collapse and expand a hierarchical graph [8], HOV-SG matches queries hierarchically [3], and BBQ shortlists objects before deductive reasoning [7].

What is missing is a controlled answer to two questions. *How much of a scene does the reasoner need?* And *does a spatial hierarchy select that evidence better than a flat cascade*—a cheap retriever that simply passes its top N records to the expensive model, the standard design in text search [10], [11]? Published systems change the hierarchy, the scorer, and

the budget at once, so their gains cannot be attributed to the hierarchy. They also rarely report how many records reach the reasoner, how many tokens it reads, or how much proxy computation remains.

This paper answers both questions under explicit controls. We formalize a recall-controlled traversal (Algorithm 1) as a strong representative of hierarchical pruning: branches are ranked by a cached proxy, kept by an adaptive probability-mass gate, and backed by a global safety shortlist that guarantees the proxy's best records reach the reasoner. We derive a cost model that states when such a cascade saves computation. We then compare the traversal, always with the same leaf scorer and the same budget, against exhaustive scoring, flat top-k, flat top-N cascades, weaker hierarchical gates, and HNSW [12], and we vary the hierarchy itself from human-authored zones to pose-only, pose+semantic, and raw RGB-D constructions.

a: Contributions

- A controlled evaluation protocol for context reduction over 3D scene records, in which the leaf scorer, the retained budget, and the query set are fixed and only candidate selection varies, with exact token accounting and paired bootstrap intervals.
- Evidence that the reasoner should read less: with Qwen2.5-7B, a BGE cascade halves input tokens and latency at a hit@3 cost whose 95% interval ends at 0.067, and three retained records yield higher answer accuracy than the full scene.
- Evidence that, at a matched budget, a spatial hierarchy does not select better evidence than a flat cascade, and that its safety shortlist makes retrieval quality insensitive to how the hierarchy is built.
- A 155,535-record scale study showing that scope, not similarity, limits flat referential retrieval, and that a language-only scene gate does not recover the scope gain.
- A release of all per-query traces, including a corrected Replica ground-truth box import and the diagnostics that located the error.

Section II reviews related work, Section III defines the traversal and the cost model, Section IV describes the data and protocol, Section V reports results, and Section VI discusses what they imply for spatial language interfaces.

II. RELATED WORK

A. LANGUAGE-ALIGNED 3D SCENE REPRESENTATIONS

A first family of methods attaches language features directly to a dense scene representation. LERF [5] distills CLIP [13] features into a neural radiance field, and LangSplat [6], Semantic Gaussians [14], and LEGS [15] do the same for 3D Gaussian splatting [16]. OpenScene [17] and ConceptFusion [2] fuse open-vocabulary features into point clouds. These representations answer a query by dense feature matching over the whole scene and do not produce discrete records that an LLM reads. Our study is complementary: it assumes that a perception stage has produced discrete records and asks how many of them the reasoning stage needs.

B. 3D SCENE GRAPHS AND HIERARCHICAL REASONING

Scene graphs make that discretization explicit. 3DSSG [18] learns object–relation graphs from indoor reconstructions; Hydra [4] builds real-time layered graphs from objects to places, rooms, and buildings; and Clio [19] adapts graph granularity to the task. ConceptGraphs [1] builds open-vocabulary object graphs from RGB-D and passes captioned objects to an LLM planner. BBQ [7], the closest object-grounding reference to our work, adds deductive LLM reasoning over a 3D object graph and reports Acc@τ on Replica and ScanNet. HOV-SG [3] and Search3D [20] organize open-vocabulary content into floor–room–object or part–object hierarchies for navigation and segmentation. SayPlan [8] is closest in spirit: an LLM collapses and expands a hierarchical 3D scene graph so that large environments fit in context. It reports the tokens saved by collapsing, but not the quality lost by pruning for a fixed scorer. None of these systems isolates the contribution of the hierarchy from that of the scorer and the budget. Table 1 summarizes the differences; numerical comparisons are restricted to same-input methods in Section V-F.

C. LANGUAGE GROUNDING BENCHMARKS IN 3D

ScanRefer [22] and ReferIt3D [23] established referring-expression grounding in ScanNet reconstructions [24], and Multi3DRefer [25] generalized it to zero or multiple targets. ScanQA and SQA3D [26], [27] extended the same scenes to question answering, while 3DRefTransformer and ScanEnts3D [28], [29] learn phrase–object correspondences. OpenLex3D [30] argues for tiered evaluation of open-vocabulary representations. These benchmarks measure whether the right object is found, not how much of the scene had to be examined. We reuse the ReferIt3D expressions and the BBQ scene subset and add that cost axis. For object maps predicted from RGB-D, OpenMask3D, Open3DIS, and Mask3D [31]–[33] are alternative instance predictors to the ConceptGraphs maps used here.

D. CONTEXT REDUCTION AND RETRIEVAL CASCADES

In text retrieval-augmented generation, prompt compression and context pruning [34]–[38] reduce the tokens given to a generator, and RAPTOR and GraphRAG [21], [39] organize corpora into trees or communities. Cascades of a cheap retriever and an expensive reranker are standard in text search, and ColBERT and PLAID [10], [11] make the efficiency–quality trade-off of late interaction explicit. HNSW and FAISS [12], [40] accelerate the first stage itself. Our traversal is a cascade whose first stage is shaped by 3D containment; the central question of this paper is whether that shaping adds anything over a flat first stage. Tool-using LLMs [41], [42] motivate our system interface, in which the traversal is exposed as a tool, but no reported number depends on a particular client.

TABLE 1. Qualitative comparison with representative systems. "LLM calls" counts calls per query at inference time. "Controlled" indicates whether the paper compares pruning against flat selection with the same scorer and budget, and reports retained records and tokens.

Method	Input	Scene structure	Query-time selection	Open vocab.	LLM calls	Controlled
ConceptGraphs [1]	posed RGB-D	object graph (flat)	full graph to LLM	yes	1	no
BBQ [7]	posed RGB-D	object graph (flat)	object shortlist + deductive LLM	yes	≥2	no
HOV-SG [3]	posed RGB-D	floor–room–object	hierarchical CLIP matching	yes	1 (parsing)	no
SayPlan [8]	given 3D scene graph	floor–room–asset	LLM expand/collapse	yes	many (iterative)	partial
RAPTOR [21]	text chunks	summary tree	embedding over tree	yes	0 (retrieval)	no
This study	posed RGB-D, object maps, expressions	zone or scene branches (manual or automatic)	proxy gate + safety shortlist vs. flat cascades	via scorer	0–1	yes

TABLE 2. Notation and terminology.

Symbol / term	Meaning
record, leaf ℓ	a view, object, or expression; the unit that is scored
branch b, C_b	a zone, region, or scene; the records it contains
N, M	number of records; number of top-level branches
$s(q,\ell)$	cheap proxy score (cached)
$r(q,\ell)$	leaf scorer (lexical, BGE, CLIP, or LLM)
$g_b(q)$, $p_b(q)$	branch score; branch probability
τ, T, B	probability-mass target, temperature, branch cap
m	size of the global safety shortlist
$C(q)$, N_q	retained candidate set; its size
k	budget: records returned after rescoring
context	serialized text read by the scorer or reasoner

III. METHOD

A. PROBLEM FORMULATION AND NOTATION

A scene is represented by a rooted hierarchy $H = (V, E)$. Its *leaves* $\mathcal{L} \subset V$, $|\mathcal{L}| = N$, are *records*: posed RGB-D *views* (internal benchmark), *object instances* (public object maps), or *expressions* (scale study). Each record carries searchable text and, where available, 3D boxes, spatial relations, and links to supporting observations. Internal nodes are *branches* (zones, regions, or scenes); $C_b \subseteq \mathcal{L}$ denotes the records under branch b, and M the number of top-level branches. Branches may overlap in the manual reference and are disjoint in automatic hierarchies.

A query q is answered by a *leaf scorer* $r(q,\ell)$ that ranks records. The *exhaustive* baseline scores all N records with r. A *candidate-selection* method passes a subset $C(q) \subseteq \mathcal{L}$ to r and keeps the top k records, the *budget*. Records in $C(q)$ are *retained*, and the text serialized from them is the *context*. Selection uses a cheap *proxy* score $s(q,\ell)$ that is precomputed or cached per record. The two flat baselines are *flat top-k*, which keeps the top k records under r, and the *flat cascade* top-N_q, which passes the top N_q records under s to r, with N_q matched per query to $|C(q)|$. Table 2 fixes this vocabulary; each term is used for one concept only.

B. RECALL-CONTROLLED TRAVERSAL

a: Branch scoring

We consider two branch scores. The *max-leaf* score ranks a branch by its best record,

$$g_b(q) = \max_{\ell \in C_b} s(q,\ell), \tag{1}$$

which avoids the bias of concatenated branch text against large branches but costs $O(N)$. The *summary* score ranks a branch by the proxy similarity between the query and a cached branch summary—the zone description for views, or the normalized mean record embedding for scenes—which costs $O(M)$. Branch scores are converted into probabilities with temperature T,

$$p_b(q) = \frac{\exp(g_b(q)/T)}{\sum_j \exp(g_j(q)/T)}. \tag{2}$$

b: Adaptive gate and safety shortlist

Branches are kept in descending p_b until their cumulative probability reaches τ or B branches are kept, so a query whose evidence is concentrated in one zone keeps one branch and an ambiguous query keeps several. Because any gate can drop the correct branch, the candidate set also contains the global top-m records under the proxy, $\mathrm{Top}_m(s)$; with $m \geq 1$ the proxy's best record always reaches the scorer. The leaf scorer then ranks $C(q)$ and the top k records are returned (Algorithm 1). Every quantity the algorithm computes is written to a per-query trace.

c: Parameter selection

All traversal parameters are selected on a frozen 50-query development split (Section IV). For the lexical scorer, development tuning selected max-leaf branch scores with $(\tau, T, B, m) = (0.7, 0.25, 5, 1)$; for BGE it selected summary branch scores with $(0.5, 0.1, 5, 1)$. The budget k is the smallest value whose paired hit@3 difference to exhaustive scoring has a 95% bootstrap lower bound of at least -0.05 on the development split, that saves at least 25% of records, and that excludes at least 20% of branches on average. For the lexical scorer this criterion selects $k = 3$. For BGE no budget meets it on the development split, where the traversal misses one query that flat search finds (lower bound -0.06); we use $k = 3$ for BGE as well and report the consequence. A

Algorithm 1 Recall-controlled query traversal

Require: hierarchy H with branches $\{C_b\}$; query q; proxy s; leaf scorer r; parameters τ, T, B, m, k
Ensure: ranked records R_k; trace
compute branch scores g_b (max-leaf or summary)
$p_b \leftarrow \mathrm{softmax}_b(g_b/T)$; sort branches by p_b
$S \leftarrow \emptyset$; $P \leftarrow 0$
for each branch b in that order **do**
  $S \leftarrow S \cup \{b\}$; $P \leftarrow P + p_b$
  if $P \geq \tau$ **or** $|S| = B$ **then break**
  end if
end for
$C \leftarrow \bigcup_{b \in S} C_b \cup \mathrm{Top}_m(s)$
$R_k \leftarrow$ top-k of C under $r(q, \cdot)$ ▷ $|C|$ scorer calls
write p_b, S, C, proxy count, context tokens, and R_k to the trace
return R_k

confidence fallback to exhaustive scoring and soft boundary membership were also evaluated on the development split and rejected (Section V-C).

C. COST MODEL

Let c_p and c_s be the costs of one proxy and one leaf-scorer evaluation. Exhaustive scoring costs Nc_s. With max-leaf scores, Algorithm 1 costs $Nc_p + O(M)$ for the proxy and gate plus $N_q c_s$ for rescoring; with summary scores and an approximate top-m shortlist, the proxy term drops to $O(M)$ plus the cost of the approximate search. The cascade reduces computation only if

$$Nc_p + N_q c_s < Nc_s \iff \frac{N_q}{N} < 1 - \frac{c_p}{c_s}. \tag{3}$$

Two consequences shape the evaluation. First, when the proxy and the leaf scorer are the same function ($s \equiv r$), rescoring adds no information: the shortlist already contains the global top-m records, and the traversal can at best reproduce flat top-k. We therefore report flat top-k as the fair baseline in that regime and treat the comparison as a correctness check. Second, the regime in which Eq. (3) holds by a wide margin is an LLM scorer, whose per-record cost exceeds that of a cached embedding lookup by orders of magnitude. There, any cascade saves computation, so the relevant baseline is the flat cascade with the same N_q; the question is whether the hierarchy chooses better records.

D. HIERARCHY CONSTRUCTION

We compare four hierarchies over the same 97 internal views. None of the automatic constructors reads the manual zones, and each writes its hierarchy to disk before the evaluator opens them; a source ledger records every file read.

a: Manual reference

Twenty-six human-authored zones over the five scenes, which may overlap, together with per-view semantic records (1,066 items in total) written by the authors.

Algorithm 2 Raw RGB-D hierarchy construction

Require: RGB frames $\{I_i\}$, depth maps $\{D_i\}$, poses $\{P_i\}$ of one scene
Ensure: disjoint branches with names; source ledger
verify that frames, depths, and poses are aligned
compute f_i (CLIP), p_i (translation), d_i (depth descriptor)
$z_i \leftarrow$ per-dimension standardization of $[f_i; p_i; d_i]$
for $K = 2, \ldots, \lfloor n/3 \rfloor$ **do**
  cluster $\{z_i\}$ into K groups; record the silhouette score
end for
keep the K with the highest silhouette score
name each cluster by the nearest place category in CLIP text space
serialize hierarchy and ledger; only then open evaluation labels

b: Pose only and pose+semantic

Views are connected when their camera translations p_i are closer than an adaptive threshold,

$$t = \alpha \cdot \mathrm{median}_i \min_{j \neq i} \|p_i - p_j\|_2, \qquad \alpha = 3.5, \tag{4}$$

and connected components form branches. Normalizing by the median nearest-neighbor spacing makes the threshold independent of capture scale and density. The multiplier α was fixed before evaluation, and Table 8 reports its effect. Pose-only reads only the camera poses. Pose+semantic additionally merges components whose authored view records share a complementary facing pair, or both a room type and a landmark; it therefore reads the authored records, but not the zones. At $\alpha = 3.5$, pose-only yields a single branch in two of the three indoor scenes and pose+semantic in all three, so both hierarchies are coarse.

c: Raw RGB-D

The strictest constructor reads only RGB images, depth maps, and poses. For each view it computes a frozen CLIP ViT-B/32 image embedding [13] $f_i \in \mathbb{R}^{512}$, the camera translation $p_i \in \mathbb{R}^3$, and a depth descriptor $d_i \in \mathbb{R}^6$ (valid-depth ratio and five depth quantiles). The concatenated features $[f_i; p_i; d_i]$ are standardized per dimension and clustered with k-means from a deterministic farthest-point initialization; the number of clusters $K \in \{2, \ldots, \lfloor n/3 \rfloor\}$ maximizes the silhouette score, where n is the number of views in the scene. Because every dimension is standardized, pose and depth contribute 9 of 521 equally weighted dimensions, and appearance dominates the grouping. Per-block weights applied before this standardization would cancel, so none are used; the constructor also enforces no minimum cluster size, and the resulting hierarchy contains nine single-view clusters (Table 8 reports both choices). Each cluster is named by the nearest of 20 place categories in CLIP text space (Appendix A). Names only label branches in traces and play no role in retrieval; both clusters of the drone flight receive the indoor label "stage". Algorithm 2 summarizes the protocol.

E. LEAF SCORERS

a: Lexical and dense scorers

The lexical scorer normalizes case, stop words, aliases, and plurals and ranks by token overlap plus overlap divided by query length, breaking ties by record ID. The dense scorer is the cosine similarity of BGE-small-en-v1.5 embeddings [43]. For both, the record proxy s is the same function as the leaf scorer r, which places them in the $s \equiv r$ regime of Section III-C.

b: Instruction-model scorer

The expensive scorer is Qwen2.5-7B-Instruct [44]; an additional Qwen2.5-1.5B-Instruct run on a CPU is reported for reference. For query q and record c the model receives

```
Judge scene-query relevance using
semantic meaning and affordances.
Answer exactly Yes or No.
Query: <q>. Candidate: <c>. Relevant?
Answer exactly Yes or No.
```

and records are ranked by the log-odds of the next token `Yes` versus `No`. All prompts of one query are scored in a single batched forward pass. In the cascades, the BGE proxy selects $C(q)$ and Qwen rescores only $C(q)$. As an ablation we also evaluate an LLM-only gate that first asks the same question about each top-level branch description, keeps the two branches with the highest score, and then scores their records; it needs two model invocations per query.

c: Image–text scorer

For hierarchy comparisons that must not depend on authored text, CLIP ViT-B/32 text embeddings score view images directly.

d: Object grounding

On object maps, a rule-based parser produces a target phrase, an optional relation (near, left/right, above/below, front/behind), and an optional anchor. The base score combines exact-label match, lexical overlap, object type, and a fixed affordance lexicon, capped at 1. A relation verified from stored links or box centers adds 0.2, and relation status breaks ties among equally scored objects. In the queries studied here every relation-bearing top candidate already reaches the cap, so relations act only as a tie-breaker, and sweeping the bonus over $\{0.2, 0.35, 0.5\}$ or the fallback threshold below over $\{0.35, 0.45, 0.55\}$ leaves Acc@0.25 unchanged. If the best score is below 0.45, the geometry is invalid, or a required relation is unresolved, the candidate set is expanded to all objects.

F. SYSTEM INTERFACE

The traversal and the constructors are exposed as tools through a Model Context Protocol (MCP) server, so that an interactive LLM client can build and query hierarchies. We used this interface for qualitative demonstrations only; every number in this paper is produced by the released scripts with the open models listed in Section VII.

IV. EXPERIMENTAL SETUP

A. INTERNAL BENCHMARK

The internal benchmark is built from five 3D Gaussian splatting reconstructions [16]: a conference hall, a museum, a theater, an outdoor street, and an aerial scene around an island (Fig. 1). The reconstructions are public scenes from the SuperSplat platform (https://superspl.at); scene titles, authors, and links are listed in the release. An operator placed 97 camera poses interactively in a browser viewer, and an RGB image and a depth map were rendered from the reconstruction at each pose, so depth is renderer-derived rather than measured by a sensor, and rendering artifacts of the reconstructions are part of the input. The authors wrote 150 queries covering object names, attributes, affordances, spatial relations, and multi-hop descriptions; 125 have a verified set of expected views and 25 are negatives whose target does not occur in the scene. The 125 positives are split once, before any tuning, into 50 development and 75 test queries, stratified by scene and query type. The queries, expected views, and negatives were written and checked within the project and have not been relabeled by independent annotators.

B. PUBLIC BENCHMARKS

Following the BBQ scene subset [7], we use eight Replica scenes (room0–2, office0–4; 575 official boxes; 48 positive and 8 negative queries) and eight ScanNet scenes (0011_00, 0030_00, 0046_00, 0086_00, 0222_00, 0378_00, 0389_00, 0435_00; 392 official boxes) with 48 Nr3D/Sr3D+ queries [23] and an extended track that adds 8 positives and 8 negatives. The 48 ScanNet queries were used while the object scorer was developed, so all ScanNet object-grounding results are in-sample. Object grounding is evaluated on *oracle maps*, which contain the official object IDs and boxes and isolate retrieval, and on *predicted maps*, produced by ConceptGraphs [1] from 3–7 sampled RGB-D frames per scene.

a: Replica ground-truth correction

Replica stores each object box in its local frame together with a rotation. Our initial importer read the local box as a world-frame box and dropped the rotation, whose per-scene median is between 12° and 76°. Rotating the boxes into the world frame raises the median best IoU between the saved predicted boxes and the ground truth from 0.065 to 0.407 and reduces the median distance from a target to the nearest predicted box of the same label from 1.56 m to 0.36 m. Feeding the ground-truth boxes through the evaluation as predictions yields Acc@0.25 and Acc@0.5 of 1.000 on both datasets. ScanNet boxes share the axis-aligned frame of the predictions, and no correction was needed there. All Replica predicted-map numbers below use the corrected boxes.

C. SCALE STUDY

We index all 155,535 Nr3D and Sr3D+ expressions from 1,436 ScanNet scenes (16,650 objects) in nested pools of 1,000, 10,000, 50,000, 100,000, and 155,535 records. A query is an expression whose own record is removed from the pool;

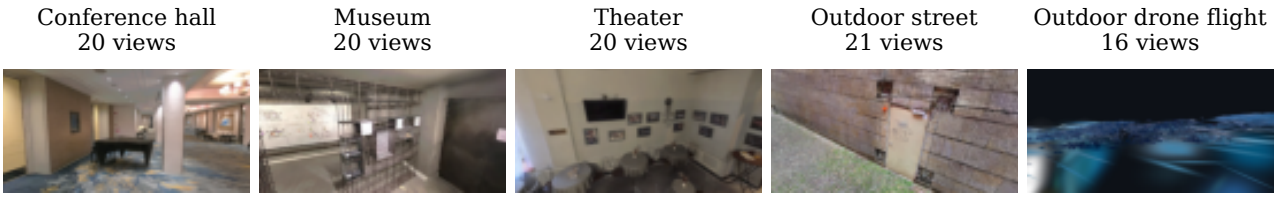


FIGURE 1. **The five scenes of the internal benchmark, rendered from 3D Gaussian splatting reconstructions, with the number of posed RGB-D views per scene.**

a hit is another expression of the same scene and object. This is expression-to-expression retrieval, a proxy for retrieving text-described object records, not 3D box grounding. Each pool evaluates up to 500 queries (88 at the 1,000 pool, where few objects have two expressions), 2,088 query–pool pairs in total. Hierarchy parameters are tuned on 2,000 further query–pool pairs that do not overlap the evaluation queries.

D. METRICS AND STATISTICS

View retrieval uses hit@1, hit@3, and mean reciprocal rank (MRR) over positive queries; for methods that return k records, MRR is computed over the returned list. Answer accuracy is the fraction of queries for which the model's chosen record is an expected view. Object grounding uses exact-ID Recall@k and Acc@τ, the fraction of positive queries whose top-ranked box has 3D IoU of at least τ with the ground truth; negatives have no box, are excluded from these denominators, and are evaluated by negative accuracy, the fraction for which the system abstains. Structural agreement with the manual zones uses macro pairwise F1 over co-clustered view pairs, reported beside a cluster-size-matched random partition and the best disjoint projection of the overlapping manual zones. Context tokens of view records are counted with the `cl100k_base` tokenizer for all non-LLM scorers and with the Qwen tokenizer for Qwen; object-map tokens are character/4 estimates. Construction cost is reported separately from query cost. Differences between methods are paired by query, with 95% intervals from 10,000 bootstrap resamples; single proportions use Wilson intervals.

V. RESULTS

A. AN LLM REASONER SHOULD READ LESS

Table 3 and Fig. 2 evaluate the regime in which Eq. (3) predicts a benefit: Qwen2.5-7B scores the records, and a BGE proxy decides which records it reads. Scoring all 19.4 records per query costs 2,236 input tokens and 5.7 s per query at hit@1/hit@3 of 0.627/0.973 on the 75 held-out positives. Algorithm 1 retains 9.48 records, halves the input to 1,090 tokens and the latency to 2.9 s, and obtains 0.653/0.947. The paired hit@3 difference is -0.027 (95% CI $[-0.067, 0.000]$) and the hit@1 difference $+0.027$ ($[-0.040, 0.107]$). Truncating the proxy ranking to three records before the LLM reads them cuts the input to 338 tokens and the latency to 0.86 s at 0.640/0.960. The LLM-only two-branch gate is the worst selector: it reads more (1,767 tokens in two calls) and loses more (hit@3 0.880), because the target falls outside its two branches for 9.3% of queries. The Qwen2.5-1.5B CPU run shows the same pattern for that gate (hit@3 0.867 versus 0.907 on the same test queries). On all 125 positives, including the development queries, the hit@3 cost of Algorithm 1 relative to exhaustive scoring is 0.032 (95% CI $[-0.064, -0.008]$), so a small recall loss is real.

A shorter context also helps when the model must commit to one answer. Given the query and a numbered list of records, Qwen2.5-7B outputs one record identifier by greedy decoding (Table 4). With decoding constrained to identifiers that occur in the list, it selects an expected record for 0.613 of the held-out queries when it reads all 19.4 records and for 0.733 when it reads the three records retained by Algorithm 1 ($+0.120$, 95% CI $[0.013, 0.227]$); flat BGE top-3 gives the same 0.733. With unconstrained decoding the gap is similar (0.587 versus 0.720), and the full list additionally produces invalid identifiers for 13% of queries. The constrained decoder was added after the unconstrained run had been inspected; both are reported. The short-context conditions forgo only the 4% of queries whose target is outside the three retained records.

B. THE HIERARCHY DOES NOT SELECT BETTER RECORDS THAN A FLAT CASCADE

At a matched budget, the hierarchy's choice of records is not better than the proxy's own ranking. In Table 3, Algorithm 1 and the flat cascade read the same number of records per query; the traversal's hit@3 is lower by 0.013 (95% CI $[-0.040, 0.000]$) and its hit@1 is identical, because its gate drops the target from $C(q)$ for one more query than flat selection. The max-leaf variant reaches the best hit@1 (0.693) but reads 42% more records than the flat cascade, so it is not a matched comparison.

When the proxy equals the scorer, Table 5 and Fig. 3 show the correctness check predicted by the cost model. At $k = 3$ the traversal returns the same top-1 record as flat top-3 and places the target at the same rank on each of the 75 held-out queries for both scorers; the full top-3 set is identical for 61 (lexical) and 53 (BGE) queries, and the remaining lists differ only in records that are not targets. Hit@1 and hit@3 are identical to flat top-k at every tested budget. Hence hit@1, hit@3, and MRR at $k = 3$ equal those of flat top-3; the MRR loss relative to exhaustive scoring (0.833 to 0.822 for lexical, 0.796 to 0.789 for BGE) is caused by truncation to three

TABLE 3. LLM leaf scorer (Qwen2.5-7B-Instruct, bfloat16, Apple M4 Pro GPU) on the 75 held-out positives. "Retained" is the number of records the LLM scores; tokens are native Qwen input tokens per query; latency is the median wall time per query including the proxy. Rows (b) and (c) retain the same number of records per query, as do rows (d) and (e). Paired differences (95% CI): (c)−(a) hit@3 −0.027 [−0.067, 0.000]; (c)−(b) hit@3 −0.013 [−0.040, 0.000] and hit@1 0.000 [−0.067, 0.067]; (e)−(d) zero for every metric.

	Candidate selection before the LLM	hit@1	hit@3	MRR	Target in C	Retained	Tokens	Latency (s)
(a)	none (all records)	0.627	0.973	0.796	1.000	19.40	2,236	5.68
(b)	flat BGE top-N_q	0.653	0.960	0.800	0.987	9.48	1,090	2.89
(c)	Algorithm 1, summary gate	0.653	0.947	0.793	0.973	9.48	1,090	2.87
(c′)	Algorithm 1, max-leaf gate	0.693	0.960	0.819	0.987	13.47	1,550	3.90
(d)	flat BGE top-3	0.640	0.960	0.784	0.960	3.00	345	0.81
(e)	Algorithm 1, $k = 3$	0.640	0.960	0.784	0.960	2.95	338	0.86
(f)	LLM-only two-branch gate	0.653	0.880	0.757	0.907	11.15	1,767	4.34

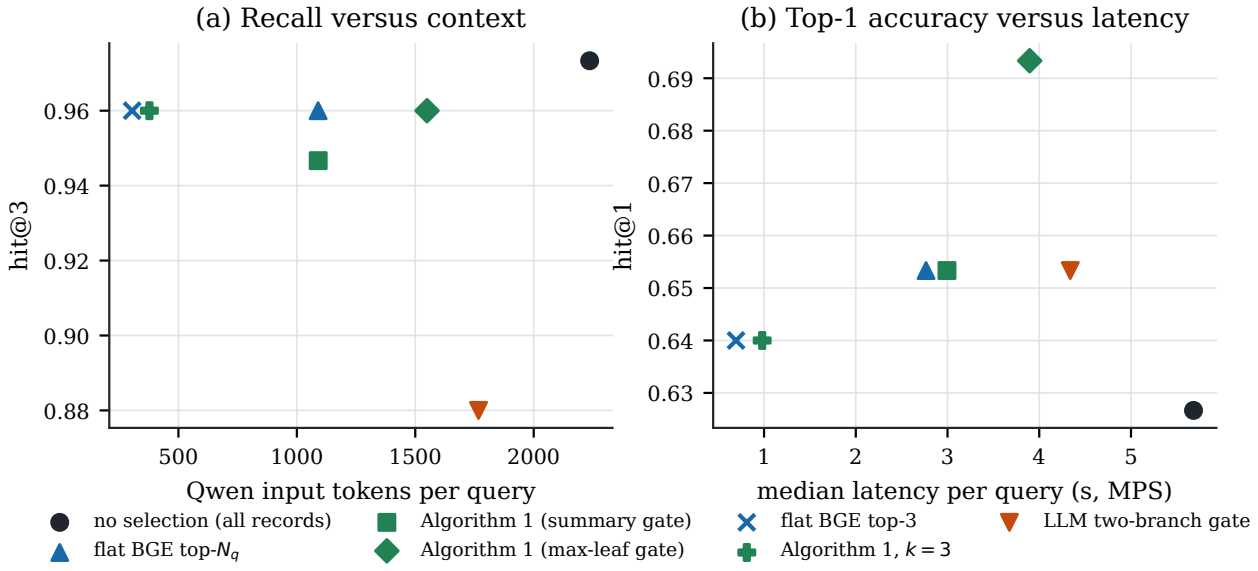


FIGURE 2. Candidate selection in front of Qwen2.5-7B on the 75 held-out positives. (a) hit@3 against the number of input tokens the LLM reads; (b) hit@1 against median latency per query. Coinciding points are offset horizontally for visibility (see Table 3 for exact values).

TABLE 4. Answer selection by Qwen2.5-7B on the 75 held-out positives: the model reads the query and a numbered list of records and outputs one record ID, with decoding constrained to listed IDs or unconstrained (free).

	Accuracy			
Context	constr.	free (valid)	Records	Tokens
all records	0.613	0.587 (0.867)	19.40	1,343
flat BGE top-3	0.733	0.707 (0.960)	3.00	266
Alg. 1, $k = 3$	0.733	0.720 (0.973)	2.95	262

records, not by the hierarchy. The traversal also serializes 125 tokens of branch summaries, so it sends more context than flat top-3 (461 versus 343 tokens for lexical, 448 versus 328 for BGE).

C. WHICH COMPONENTS OF THE TRAVERSAL MATTER

Figure 4 adds the components of Algorithm 1 one at a time to a fixed two-branch gate at $k = 3$. On the development split, lexical hit@3 rises from 0.700 to 0.740 with adaptive mass, to 0.840 with max-leaf branch scores, and to 0.860 with the safety shortlist; BGE rises from 0.780 to 0.860, almost entirely through the shortlist (for BGE the branch score is the dev-selected summary score). On the held-out split the components act in the same way (lexical 0.787 to 0.933, BGE 0.813 to 0.960); there adaptive mass gives the largest lexical gain, and for BGE the configured branch score adds nothing on either split. The confidence fallback did not improve the development criterion and was disabled, so the selected configuration never falls back; soft boundary membership was rejected for the same reason. The components that matter are therefore the ones that make the traversal behave like flat search: scoring branches by their best record and guaranteeing that the best record survives.

The traversal parameters behave the same way (Fig. 5). On the development split, hit@3 is unchanged across τ, T, and the boundary width, and drops only for a branch cap of one (0.84) or without the shortlist ($m = 0$, 0.84). The budget k is the only parameter that trades quality for records. With $s \equiv r$ and $m \geq 1$, a target ranked first by the scorer always survives, and targets ranked second or third are lost only if

TABLE 5. Proxy equal to scorer on the 75 held-out positives. Proxy comparisons are 19.4 records per query in every row. Context tokens use `cl100k_base`; the traversal's context includes 125 tokens of branch summaries.

Scorer	Candidate selection	hit@1	hit@3	MRR	Retained	Context tokens	vs. exhaustive	Same top-3 set
Lexical	exhaustive	0.733	0.933	0.833	19.40	2,153	–	–
Lexical	flat top-3	0.733	0.933	0.822	3.00	343	−84.1%	–
Lexical	Algorithm 1, $k = 3$	0.733	0.933	0.822	2.99	461	−78.6%	61/75
BGE	exhaustive	0.667	0.960	0.796	19.40	2,153	–	–
BGE	flat top-3	0.667	0.960	0.789	3.00	328	−84.7%	–
BGE	Algorithm 1, $k = 3$	0.667	0.960	0.789	2.95	448	−79.2%	53/75

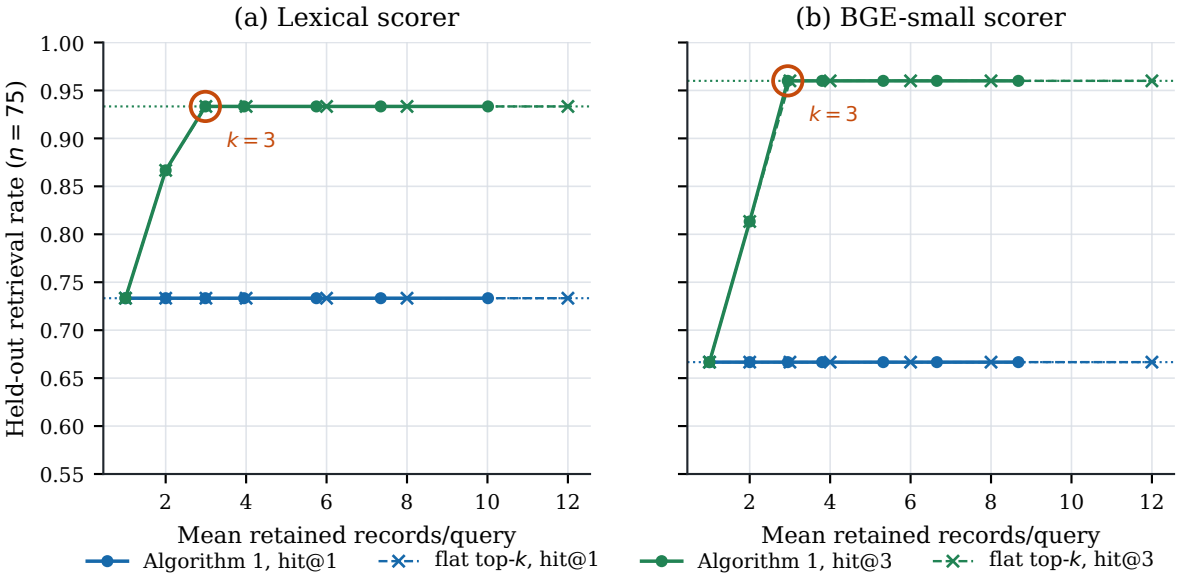


FIGURE 3. Held-out quality versus retained records ($n = 75$) as the budget k varies. Algorithm 1 (solid) and flat top-k under the same scorer (dashed) have identical hit@1 and hit@3 at every budget; dotted lines are exhaustive scoring.

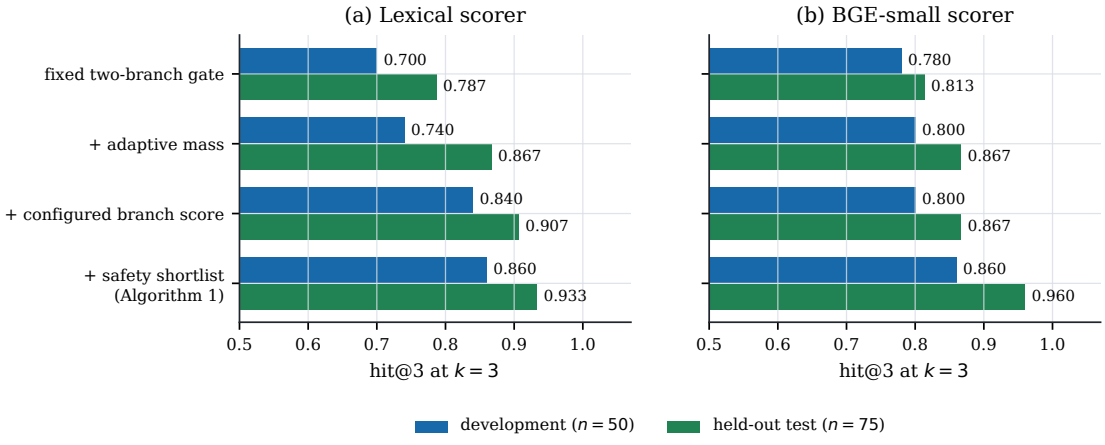


FIGURE 4. Component ablation of Algorithm 1 at $k = 3$ on the development split, where the configuration was chosen, and on the held-out split. Each row adds one component to a fixed two-branch gate.

they lie outside the kept branches, so the gate parameters seldom matter.

All five held-out lexical misses (Table 6) occur outdoors. Because the traversal ranks every target exactly as flat top-

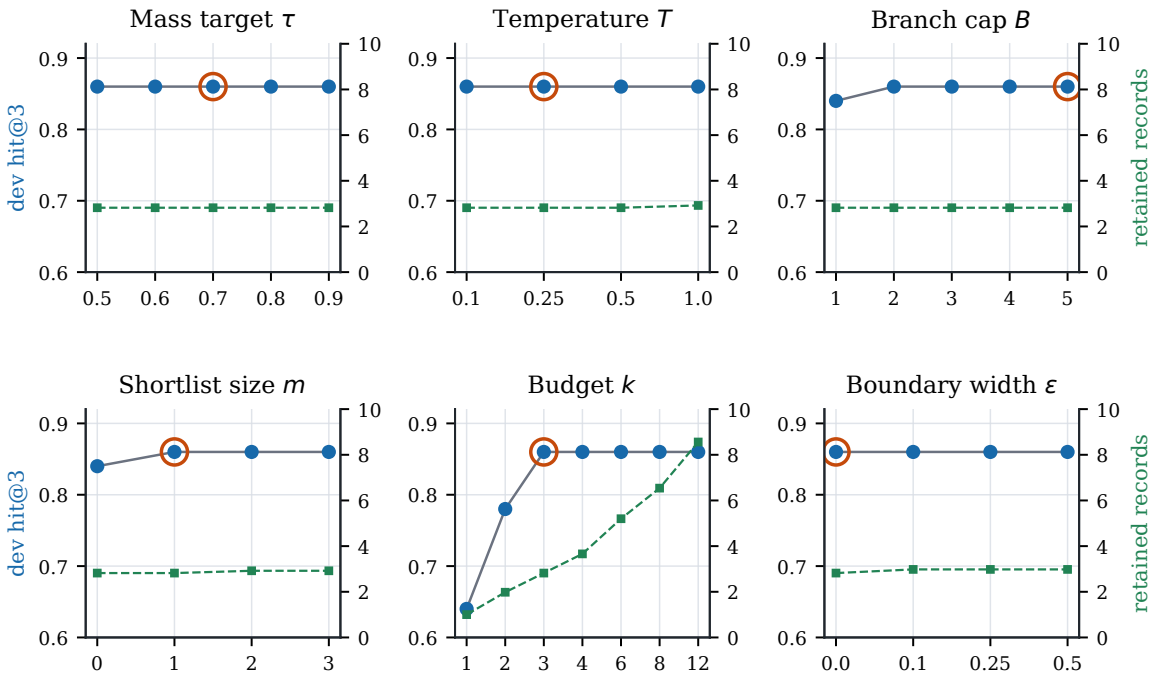


FIGURE 5. One-at-a-time sensitivity of development hit@3 (blue, left axis) and retained records (green, right axis) to the traversal parameters (lexical scorer, $n = 50$). Rings mark the selected values.

TABLE 6. All held-out lexical hit@3 misses ($n = 75$). "Oracle scope" gives hit@3 when search is restricted to the zone that contains the target.

Scene	Query (abridged)	Oracle scope
street	glass-block wall on the right	no
street	move between street levels	yes
street	utility access area	yes
street	street wayfinding marker	no
drone	rocky island surrounded by water	no

3 does, all five are failures of the scorer. For two of them, restricting search to the zone that contains the target (an oracle scope) recovers it, which shows that the correct zone would have removed a distractor; the language-driven gate does not find that zone.

a: Negative queries

An abstention rule set on the development positives—abstain when the best proxy score is at or below the fifth percentile of the development positives' best scores—abstains on all 25 internal negatives with BGE, while answering 73 of the 75 held-out positives (hit@3 0.933 with abstentions counted as misses). The lexical scorer abstains only when no token overlaps; it abstains on 22 of 25 negatives and answers every positive. Because the shortlist keeps the global best record, flat search and the traversal make identical abstention decisions.

D. THE SAFETY SHORTLIST MAKES THE HIERARCHY INTERCHANGEABLE

Table 7 compares the four hierarchies on all 125 positives under three leaf scorers and two gates at $k = 3$. Under Algorithm 1, no hierarchy differs from the human zones in hit@1 or hit@3 under any scorer (every paired interval contains zero), and none differs from flat search either: the largest gaps, -0.032 for raw RGB-D under CLIP ($[-0.088, 0.024]$) and -0.024 under BGE ($[-0.056, 0.000]$), are within the resolution of the benchmark. The grid uses max-leaf branch scores for every scorer and an untuned CLIP temperature of 0.01. Lexical hit@3 is 0.904 for every hierarchy, equal to flat, and the values do not change at $k = 6$ or $k = 12$. Under a weaker budget-filling gate, which orders branches by max-leaf score but has no mass cut and no shortlist, point estimates spread further—the human zones reach 0.864 under lexical and 0.552 under CLIP scoring, and raw RGB-D 0.848 under BGE—while the coarse pose hierarchies stay close to flat because they are nearly flat; at $k = 3$ none of these differences is significant at $n = 125$.

A fixed-width gate magnifies the dependence on the hierarchy. When only the two best branches are kept and there is no shortlist (Fig. 6), the target lies outside the kept branches for 21 of 125 queries with the human zones, for 29 with raw RGB-D clusters, and for 2 with either pose hierarchy. CLIP hit@3 is 0.624 for flat search, 0.592 for the human zones, and 0.528 for raw clusters; raw clusters are significantly worse than flat search (-0.096, 95% CI $[-0.168, -0.032]$) but not significantly worse than the human zones (-0.064, $[-0.136, 0.008]$). Replacing the fixed cut by Algorithm 1

TABLE 7. Hierarchy–scorer grid on 125 positives: hit@3 at $k = 3$ under Algorithm 1 and under a budget-filling gate without mass cut or shortlist. "Br." is the number of top-level branches over the five scenes.

		Algorithm 1			Budget-filling gate		
Hierarchy	Br.	Lex.	BGE	CLIP	Lex.	BGE	CLIP
Flat (no hierarchy)	–	0.904	0.928	0.624	0.904	0.928	0.624
Manual zones	26	0.904	0.920	0.600	0.864	0.880	0.552
Pose only	10	0.904	0.920	0.624	0.888	0.880	0.624
Pose+semantic	8	0.904	0.920	0.624	0.920	0.896	0.624
Raw RGB-D	27	0.904	0.904	0.592	0.880	0.848	0.592

closes the gap to flat search for every hierarchy within the resolution of the benchmark.

a: Construction cost and structure

Pose-only construction for all five scenes takes 0.8 ms, pose+semantic 6.3 ms, and raw RGB-D construction a median of 17 s (12–23 s over three runs on a shared CPU; 97 images). The automatic hierarchies agree with the human zones only moderately (Fig. 7): macro pairwise F1 is 0.588 for pose only, 0.570 for pose+semantic, and 0.487 for raw RGB-D, against size-matched random partitions of 0.541, 0.563, and 0.366 and a best attainable disjoint score of 0.877. The pose scores are close to random because a single-branch scene scores the same as its random baseline. Table 8 shows that the construction constants change structure more than retrieval: under CLIP and the budget-filling gate, hit@3 stays within 0.568–0.632 over all tested values, while pairwise F1 ranges from 0.314 to 0.600.

E. SCOPE, NOT SIMILARITY, LIMITS RETRIEVAL AT SCALE

Figure 8 measures expression retrieval on pools of up to 155,535 records. At the largest pool, exact dense BGE retrieval obtains hit@3 0.236 and HNSW [12] 0.228; restricting search to the correct scene—an oracle that assumes the scan ID is known—raises hit@3 to 0.948 while scoring 284 records. Of the flat misses at this pool, 96.9% have all three returned expressions in other scenes (96.6% over all pools): the difficulty of this task is scope.

A language-only scene hierarchy does not recover that gain. Algorithm 1 over scene branches, tuned on disjoint queries, at best matches flat search. With max-leaf branch scores it returns the flat result (hit@3 0.234 in the same run as flat 0.234) but still scores every record. With scene-summary scores and a three-record HNSW shortlist it obtains 0.232 (paired difference -0.002, 95% CI $[-0.012, 0.008]$) while computing about 2,080 vector similarities instead of 155,535. Without the shortlist its hit@3 falls to 0.116, and at the largest pool its gate selects the query's own scene for only 3–27% of queries, depending on the configuration. The retained accuracy therefore comes from the approximate shortlist, not from the scene gate. HNSW alone obtains 0.228 in a separate run with about 584 distance computations. The summary configuration with a single branch and a three-record shortlist was added to the tuning grid after inspecting tuning results, but before any evaluation query was scored. Flat hit@3 at the largest pool varies between 0.234 and 0.242 across runs because many Sr3D+ template expressions tie exactly.

F. OBJECT GROUNDING

Table 9 reports object grounding. On oracle ScanNet maps, Algorithm 1 with the object scorer reaches Recall@1 and Acc@0.25 of 0.604, matching flat lexical Acc@0.25 (0.604; flat Recall@1 0.625), while scoring 21.3 instead of 49.0 objects per query; Replica is a ceiling at 1.000 for every method in Table 9. On predicted maps, accuracy is bounded by perception. On ScanNet, flat and hierarchical BGE both place a box with IoU$\geq$0.25 for 6 of 48 queries, whereas the same scorer reaches 17 of 48 (0.354) on the oracle map, and even the top three BGE candidates contain a box with IoU$\geq$0.25 for only 9 of 48 queries. With corrected ground truth, native ConceptGraphs reaches Acc@0.25 of 0.271 on Replica (0.063 before the correction). The paired BGE runs on Replica predicted maps could not be recomputed from the saved outputs and are omitted. On the 8 negatives per track, lexical variants abstain because no token overlaps (negative accuracy 1.000); embedding scorers and native ConceptGraphs never abstain (0.000).

LangSplat [6] was run once, on ScanNet 0011_00 with six queries, 3,000 instead of 30,000 optimization iterations, and a 6 GB GPU; it selected an expected view for 4 of 6 queries and outputs points rather than boxes. We report it only as evidence that the pipeline runs on our data. We did not run BBQ, whose released evaluation uses a different map and LLM protocol, so its published numbers are not comparable with ours.

VI. DISCUSSION

A. WHAT THE RESULTS SAY ABOUT SPATIAL LANGUAGE INTERFACES

a: The reasoner should read little, and a flat cheap-first cascade decides what well

The clearest result is that the expensive model does not need the scene. With a 7B instruction model, a cached BGE ranking that passes about half of the records halves tokens and latency at a hit@3 cost whose interval ends at 0.067, and a three-record context raises the model's answer accuracy by 0.12. The limiting factor is the recall of the cheap stage, which is 0.96 at three records on this benchmark. Designers of spatial language interfaces can therefore spend their effort on the first-stage ranking and on the budget rather than on the reasoner's context window.

b: Spatial hierarchy is a representation, not a retrieval gain

We found no setting in which the hierarchy selected better evidence than the flat ranking of the same proxy at the same budget: not with lexical, dense, image–text, or LLM scorers, not with human or automatic zones, and not at 155,535 records. The components that make the traversal competitive—max-leaf scoring and the safety shortlist—are exactly those that make it reproduce the flat ranking, and the one component

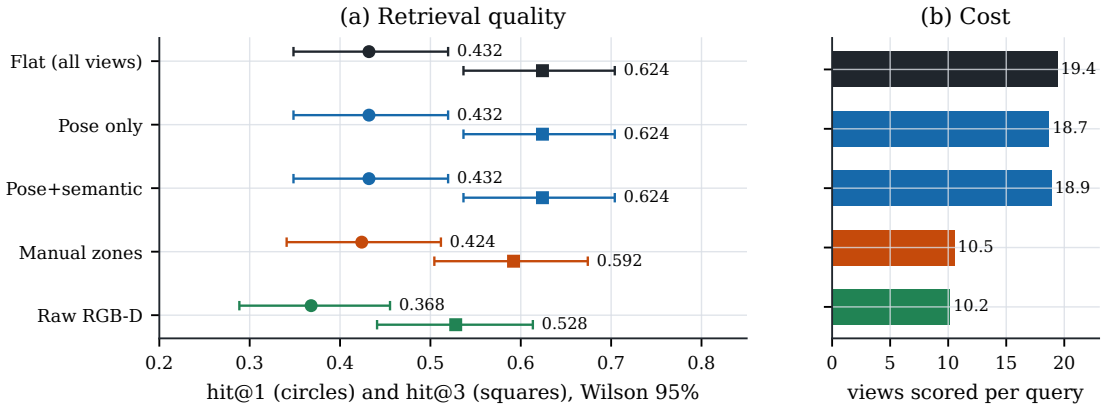


FIGURE 6. **Fixed two-branch gate with the CLIP scorer (125 positives), an ablation that Algorithm 1 replaces. (a) hit@1 and hit@3 with Wilson 95% intervals; (b) views scored per query.**

TABLE 8. **Construction constants (125 positives; CLIP scorer, budget-filling gate, $k = 3$). Defaults in bold. The minimum-cluster rule is not part of the released constructor and was added for this sweep.**

Hierarchy	Constant	Values	F1	hit@3
Pose only	α	2, 2.5, 3, **3.5**, 4, 5	0.475–0.600	0.576–0.624
Raw RGB-D	$K_{\max}$	$n/4$, $\boldsymbol{n/3}$, $n/2$	0.496, 0.487, 0.314	0.608, 0.592, 0.568
Raw RGB-D	min. views	**none**, 2, 3, 4	0.487–0.545	0.592–0.632
Raw RGB-D	pose weight	0, 0.25, **0.5**, 1	0.487–0.497	0.592
Raw RGB-D	depth weight	0, 0.1, **0.2**, 0.5	0.487–0.499	0.592–0.608

TABLE 9. **Object grounding over positive queries. Objects: mean objects scored per query. ScanNet queries were used during scorer development (in-sample). Predicted maps are ConceptGraphs maps; Replica uses the corrected ground truth. *Spatial median-split hierarchy with $\tau = 0.7$, $m = 1$; it scores all cached proxies and keeps 3 objects.**

Map	Dataset (n)	Method	R@1	Acc@0.25	Acc@0.5	Objects
oracle	ScanNet (48)	flat lexical + relations	0.625	0.604	–	49.0
oracle	ScanNet (48)	Algorithm 1 + relations	0.604	0.604	–	21.3
oracle	ScanNet (48)	flat BGE	0.333	0.354	0.333	49.0
oracle	Replica (48)	all methods	1.000	1.000	1.000	30.3–71.9
predicted	ScanNet (48)	flat BGE	–	0.125	0.021	50.4
predicted	ScanNet (48)	hierarchical BGE*	–	0.125	0.021	50.4
predicted	ScanNet (48)	native ConceptGraphs	–	0.063	0.000	–
predicted	Replica (48)	native ConceptGraphs	–	0.271	0.146	–

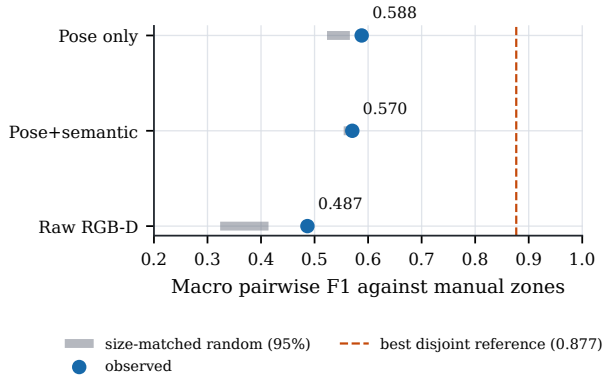


FIGURE 7. **Structural agreement (macro pairwise F1) of automatic hierarchies with the human zones, beside cluster-size-matched random partitions and the best disjoint projection of the overlapping human zones.**

that could add scope, the language-driven gate, chose the correct scene for at most 27% of queries at the largest pool. A hierarchy should therefore be kept for what it provides independently of ranking: a traceable route, an interface for spatial scoping by an external signal, and structure for summaries. If a hierarchy is used for pruning, the shortlist is not optional: without it, a gate loses up to a tenth of hit@3.

c: Scope is valuable when it comes from outside the query

The scale study isolates a failure mode of flat retrieval that grows with the collection: short descriptions are equally plausible in many scenes, and almost every miss is an other-scene record. Knowing the scene lifts hit@3 from 0.236 to 0.948, whereas inferring it from the query text does not. Scope should therefore come from localization, user context, or the robot's position, and the hierarchy is the natural place to apply

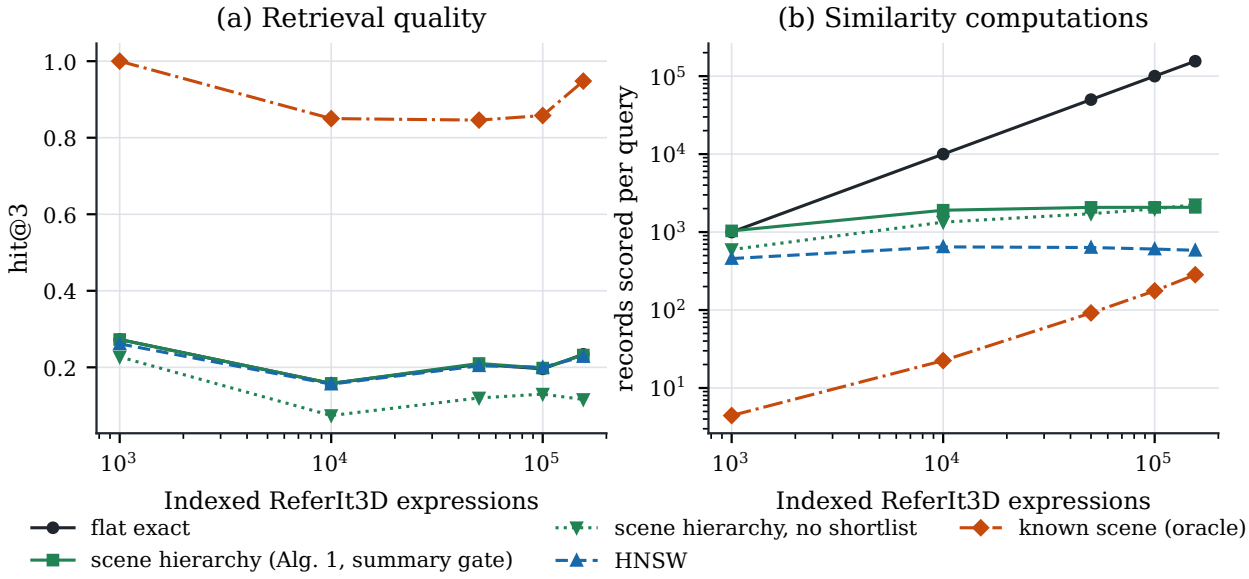


FIGURE 8. **Expression retrieval over nested ReferIt3D pools (query expression removed; a hit is another expression of the same scene and object). (a) hit@3; (b) similarity computations per query (for HNSW: computed distances; for the scene hierarchy: branch summaries, HNSW distances, and rescored records). The known-scene control assumes the scan ID is known.**

it.

d: Human zones are not needed for pruning

Under the proposed traversal, pose-only hierarchies built in under a millisecond and raw RGB-D clusters were interchangeable with human-authored zones. Because the shortlist removes the dependence on branch quality, manual zoning can be skipped.

e: Perception and evaluation dominate end-to-end grounding

On predicted maps, every retrieval method was bounded by the map, and a single unrotated box import had understated Replica accuracy about fourfold. Retrieval methods for 3D grounding should report oracle- and predicted-map results with the same scorer and verify their ground truth by passing it through the evaluator.

B. THREATS TO VALIDITY

Internal validity. Traversal parameters were selected on a development split frozen before tuning, and all comparisons are paired, but the test split has 75 positives, so differences of a few points cannot be resolved. The internal queries and labels were written by the authors, and the ScanNet object queries were used while the object scorer was developed. *Construct validity.* Retained records and tokens measure what the reasoner reads, not monetary cost; latencies were measured on one consumer GPU without a serving framework, while other experiments shared the machine, so only their ratios are informative. *External validity.* The internal benchmark has five scenes rendered from reconstructions, with renderer-derived depth, the scale study uses expressions rather than boxes, the predicted-map results rest on one perception system with sparse frames, and the largest reasoner has 7B parameters. The conclusion that hierarchy does not beat a flat cascade holds for the scorers and budgets tested; a proxy that encodes geometry the leaf scorer lacks could behave differently.

VII. REPRODUCIBILITY AND DATA AVAILABILITY

Code, the 150 queries with expected views and splits, the authored view records and camera poses, per-query traces (branch probabilities, retained IDs, token counts, latencies, and model scores), the corrected Replica ground-truth import, and the scripts that regenerate every figure and table are released. The rendered RGB-D views are not redistributed (see Ethics and Consent). The ReferIt3D, ScanNet, and Replica data are available from their original providers under their licenses; we release only derived records and identifiers. The models are `Qwen/Qwen2.5-7B-Instruct` revision `a09a35458c702b33eeacc393d103063234e8bc28` (bfloat16), the saved `Qwen/Qwen2.5-1.5B-Instruct` revision `989aa7980e4cf806f80c7fef2b1adb7bc71aa306` runs, `BAAI/bge-small-en-v1.5` (FastEmbed ONNX export), and `openai/clip-vit-base-patch32`. The release is available at https://github.com/LeoVesinML/beyond-proximity-release. LLM experiments ran on an Apple M4 Pro (14 CPU cores, 48 GB unified memory) using its GPU; the saved 1.5B runs used a CPU. Tokens of non-LLM contexts are counted with `tiktoken cl100k_base`.

ETHICS AND CONSENT

The internal benchmark contains no images of people captured by the authors; all views are renders of existing 3D

Gaussian splatting reconstructions. We inspected every released view: none shows an identifiable person, and the only vehicle, in one street view, has no legible license plate. Some views show framed photographs and exhibition panels that are part of the reconstructed interiors. The reconstructions were published by their authors on the SuperSplat platform, whose terms of use allow others to use, reproduce, and create derivative works from publicly posted content with attribution to its authors; we credit each scene's author in the release. We do not redistribute the reconstructions or the rendered views; the released records, poses, and results contain no image content.

VIII. CONCLUSION

We asked how much of a 3D scene a language model must read and whether a spatial hierarchy chooses that evidence better than a flat cascade. Under same-scorer, matched-budget controls, the answer to the first question is "very little": a cheap proxy that passes half of the records preserves most of the reasoner's recall while halving its cost, and a three-record context improves its answers. The answer to the second is "no": the recall-controlled traversal ranks every target as the flat ranking of its proxy does, its safety shortlist makes the hierarchy interchangeable, and a language-driven scene gate does not recover the large scope gain that external localization provides at 155,535 records. Hierarchies remain useful as an interface for externally supplied scope and for traceable routes, and pose-based hierarchies provide them without annotation.

ACKNOWLEDGMENT

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APPENDIX A PLACE VOCABULARY OF THE RAW RGB-D CONSTRUCTOR

Clusters are named with the CLIP prompt "a photograph of a {label}" and the following 20 labels: auditorium, banquet hall, conference room, corridor, entrance, exhibition gallery, facade, garden, information desk, lobby, lounge, museum display, outdoor plaza, reception desk, service counter, stage, street, theater seating, walkway, and wayfinding area. Five of them (facade, garden, outdoor plaza, street, walkway) describe outdoor places.

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